# Two-Stage Retrieval for RAG: Bi-Encoder vs. Cross-Encoder Re-Ranking

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**Code:** https://github.com/FavianaNoronha/RAG-assignment

## 1. Introduction and Setup

RAG stands for retrieval-augmented generation. In RAG, the LLM does not answer from its own memory; instead, a search step first finds the right pages in a document, and the LLM answers from those pages, much like an open-book exam. In this project, the document is the ICD-10-CM Official Guidelines FY 2026 PDF.

The LLM only gets 3 small pieces of the document, called chunks. If the right chunk is not among those 3, the LLM either says it does not know or gives a wrong answer with confidence. In this project, we compared two ways of picking those 3 chunks:

- **Pipeline 1:** a fast search that grabs the top 10 most similar chunks and passes the first 3.
- **Pipeline 2:** the same top 10, but a second, smarter model re-reads all 10 against the question and re-orders them before picking 3.

| Item | Choice |
|---|---|
| Document | ICD-10-CM Official Guidelines FY 2026 (CMS/NCHS), pp. 7–121 |
| Chunks | 601 page-bounded chunks, ~120 words, 25-word overlap (same chunks for both pipelines) |
| Pipeline 1 | all-MiniLM-L6-v2 bi-encoder, exact cosine search → top 10 → first 3 to the LLM (Answer 1) |
| Pipeline 2 | same top 10 → ms-marco-MiniLM-L-6-v2 cross-encoder → re-sorted top 3 to the LLM (Answer 2) |
| LLM | Qwen2.5-7B-Instruct (4-bit, local), greedy decoding, identical prompt for both answers |
| Test set | 19 queries with hand-written answers; rescues found by scanning ~100 paraphrased queries |

## 2. Architecture Comparison

**Bi-encoder (Pipeline 1):** It reads the question and converts it into a list of numbers called a vector. Separately, it reads each chunk on its own and converts it into a vector as well. It then compares them using cosine similarity. This is fast because the chunks never change, so all 601 chunk vectors can be computed once, in advance. When a question is asked, only the question needs encoding, followed by a quick number comparison.

**Cross-encoder (Pipeline 2):** This is the smarter but slower model. It reads the question and the chunk together, as one piece of text. Every word of the question can look at every word of the chunk (joint self-attention), so it understands phrases like "should not be assigned" or "except when". It is slow because nothing can be prepared in advance: every question–chunk pair needs a full run of the model, so it is only practical on a few candidates, not the whole document.

## 3. Rank Shifts and Answer 1 vs. Answer 2

**Table 1.** Rank of the chunk that holds the answer (top 3 = what the LLM sees), and each answer judged against the document.

| # | Type | Query (short) | Gold chunk: baseline → re-ranked | Answer 1 | Answer 2 |
|---|---|---|---|---|---|
| 1 | Exception | Burn codes T31/T32 for sequelae? | #389: 9 → 1 | No answer | No answer |
| 2 | Exception | Perinatal code for an adult? | #341: 5 → 1 | No answer | **Correct** |
| 3 | Exception | Two drugs in a poisoning? | #394: 4 → 1 | Wrong | Wrong |
| 4 | Distractor | Diagnosed HIV later asymptomatic? | #85: 9 → 1 | Wrong | **Correct** |
| 5 | Distractor | Routine signs get extra codes? | #360: 10 → 1 | Correct | Correct |
| 6 | Negation | Codes not allowed after hurricane collapse? | #72: 4 → 1 | No answer | **Correct** |
| 7 | Negation | O85 for sepsis after OB procedure? | #321: 8 → 1 | Wrong | **Correct** |
| 8 | Keyword trap | 7th char. A, fracture from disease? | #284: 8 → 2 | No answer | **Correct** |
| 9 | Keyword trap | Kidney donor visit? | #493: **1** → **10** | Correct | No answer |
| 10 | Keyword trap | Old heart attack code? | #251: **2** → **6** | Correct | No answer |
| 11 | Keyword trap | Outpatient "probable" diagnosis? | not in top 10 | Wrong | No answer |

| | | | | | |
|---|---|---|---|---|---|
| 12 | Exception | Two Excludes1 codes together? | #14: 5 → 4 | Wrong | Wrong |
| 13–19 | Controls | sepsis, Z79.4, COVID, MRSA, alcohol, etc. | already top 3 in both | Correct* | Correct* |

*Q13 (non-provider clinicians, #55: 5 → 6) is partial in both. **Totals:** Answer 1 correct 9/19, Answer 2 correct 12/19; on the 8 rescues (Q1–8), Answer 1 correct 1/8 and Answer 2 correct 6/8.

Re-ranking improved the results in Q7 and Q4. In Q7, the correct rule about the O85 sepsis code was in chunk #321, but the initial search ranked it eighth, so the LLM missed it and incorrectly stated that O85 should be assigned. After re-ranking, the chunk moved to rank 1, and Answer 2 correctly stated that the code should not be assigned. Similarly, in Q4, the initial search retrieved chunks with similar wording but the wrong meaning, leading to an incorrect answer about asymptomatic HIV coding. Re-ranking moved the correct chunk (#85) from rank 9 to rank 1, allowing Answer 2 to correctly explain that a patient with a previously diagnosed HIV illness cannot later be coded as asymptomatic HIV. These cases show how re-ranking can help distinguish between similar-looking rules with different meanings.

However, re-ranking did not improve every result. In Q12, the exception to the Excludes1 rule remained outside the top three, so both answers missed an important condition. In Q9–10, re-ranking actually pushed the correct chunk down; for the kidney donor question it fell from rank 1 to rank 10. In Q11, the relevant chunk was not in the top 10, so re-ranking could not retrieve it. Finally, in Q1 and Q3, the LLM still failed even when the correct chunk ranked first. This shows that better retrieval alone does not guarantee a correct answer.

## 4. Latency Profiling

**Table 2.** Retrieval time per query (190 timed runs on an Apple M4 GPU; LLM time excluded, as it is the same for both).

| Pipeline | Mean (ms) | p95 (ms) |
|---|---|---|
| 1: Bi-encoder (embed + search) | 5.3 | 5.8 |
| 2: Bi-encoder + cross-encoder re-rank | 41.8 | 52.6 |
| Re-rank step only (10 pairs) | 36.5 | 47.3 |

The fast search takes around 5 ms per question, while adding re-ranking increases it to 42 ms, making it roughly eight times slower. This extra time mainly comes from the re-ranking step, which takes about 36 ms. The reason is that the fast search compares the question with chunks that are already prepared, while re-ranking runs the model separately for each of the 10 question-and-chunk pairs. In practice, this delay is hardly noticeable for a single user: since the LLM takes a few seconds to generate an answer, the extra 0.04 seconds is a small price to pay for better results.

## 5. Scaling Question

**Table 3.** Measured cross-encoder cost ≈ 4 ms per (query, chunk) pair, projected to scoring every chunk for one query.

| Corpus size | 601 (this doc) | 10,000 | 1,000,000 | 10,000,000 |
|---|---|---|---|---|
| Time per query | 2.5 s | 0.7 min | 1.1 h | 11.1 h |

- The cross-encoder compares the question with every chunk separately, so the more chunks there are, the longer it takes. This process must be repeated for every new question.
- Based on Table 3, processing 601 chunks takes about 2.5 seconds per question. With 1 million chunks, it could take around an hour, making it impractical for large systems.
- A bi-encoder avoids this problem by preparing chunk vectors in advance, so it only needs to encode the question and quickly compare the numbers.
- **The best solution is a two-stage approach:** first, use the bi-encoder to shortlist the top 10 chunks, then use the cross-encoder to rank them more accurately. In our experiment, this second step took only about 36 ms.

## 6. Limitations and Conclusion

- Overall, re-ranking helped when the correct information was buried lower in the search results. Out of the eight questions where the right chunk moved to the top 3, Answer 2 got six correct, while Answer 1 got only one. It did make the search slower by around 36 ms, but compared to the few seconds taken by the LLM, this extra time is not much.

- There were some limitations in this experiment. Only 19 questions were tested, so we can't say if the same results will happen with more questions. Re-ranking also made some results worse, and it couldn't help when the correct chunk was not in the top 10. In some cases, even after finding the right information, the LLM still gave a wrong answer.I judged the answers myself, and the LLM is a small local model.

- For future work, a re-ranker trained on medical coding data could be tried to see if it understands the coding rules better.

## References


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